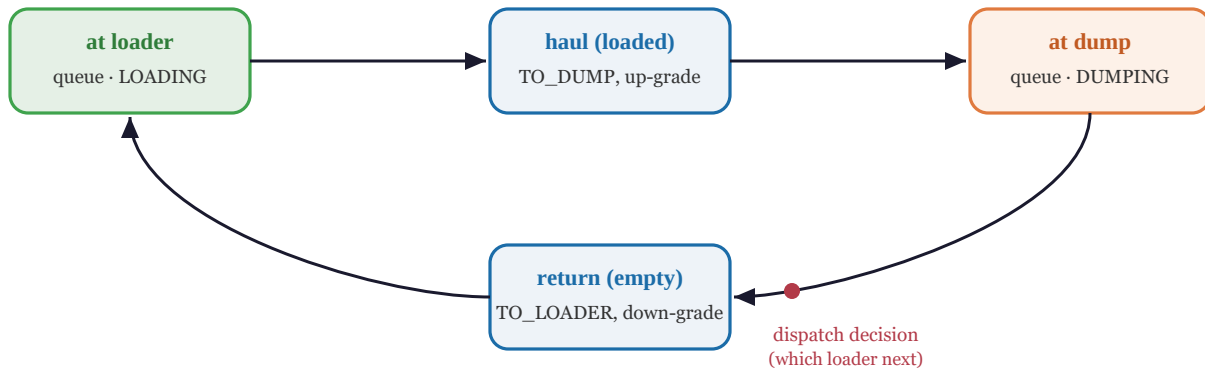


The haul cycle as a deterministic discrete-event process



travel time = $f(\text{constrained network})$:

one-way ramps · single-lane zones + passing bays ·
junction blocking · headway capacity, FIFO no-overtake
so bunching EMERGES, it is not sampled from a distribution.

speed by grade (rimpull / retarder):

attainable v solves $F(v) = \min(F_{\text{traction}}, \eta P/v)$
against grade + rolling resistance, per truck class + GVW.
Deterministic: output is a pure function of (spec, policy, seed).